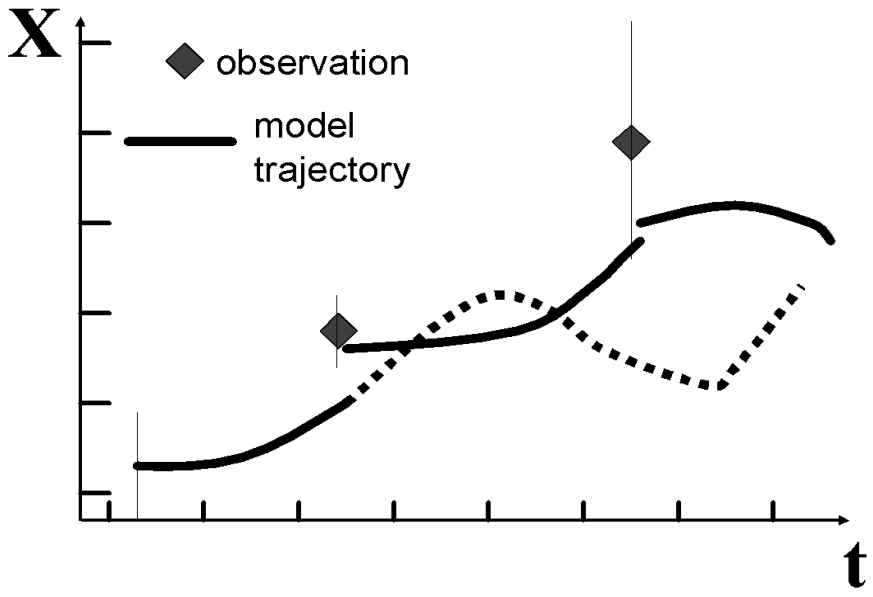


LIS-DA Overview

Yeosang Yoon
NASA GSFC/SAIC

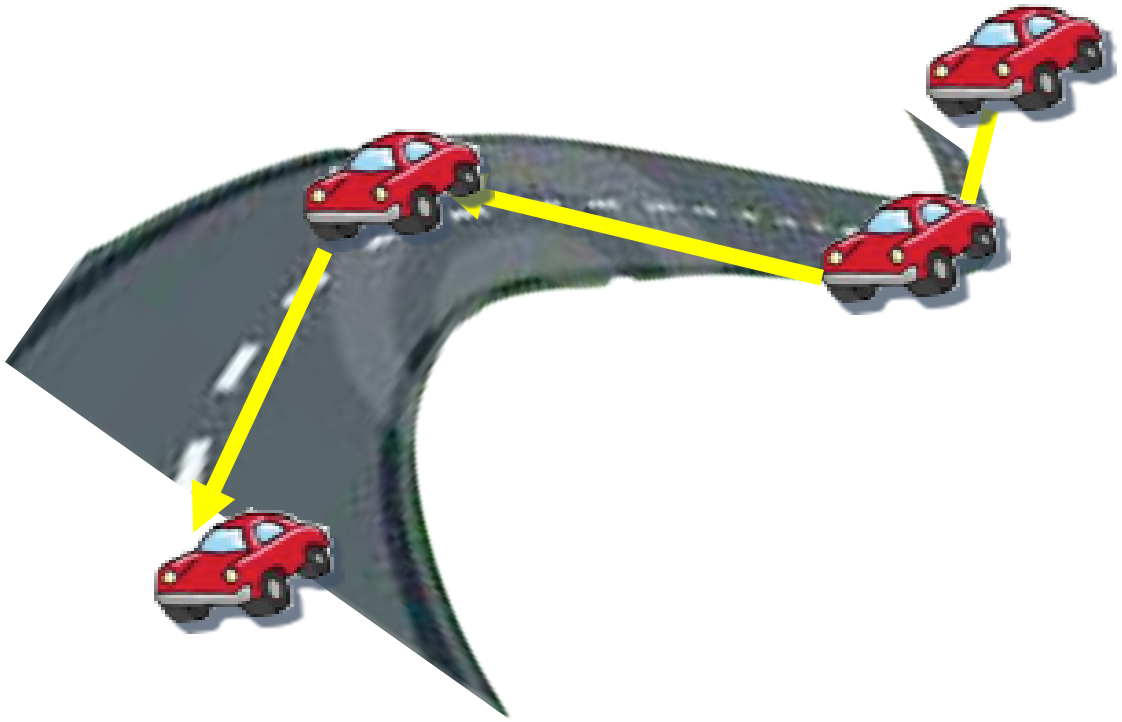
What is data assimilation?

Data assimilation is the technique whereby observational data are combined with output from a numerical model to produce an optimal estimate of the evolving state of the system.



A crude analogy

"Sleepy-driver" scenario



Why data assimilation?

- **Coverage:** DA helps in extending the spatial and temporal coverage of the remote sensing measurements. (SMAP ~3 days, Landsat ~16 days, ...)
- **Observability:** DA helps in spreading information from observations to all model variables that are connected. (e.g., Rootzone SM)
- **Resolution:** DA essentially helps in spatial and temporal downscaling and upscaling of data.
- **Data volume and redundancy:** DA can potentially filter out the redundant information in multiple datasets
- **Maximize information:** Data assimilation provides temporally and spatially complete and optimal estimates based on all available information

Types of Data Assimilation

Sequential (Filtering)

Only considers observations made in the past until the time of analysis (real-time assimilation systems).

Non-sequential (Smoothing)

Observations from the future can be used (e.g. reanalysis).

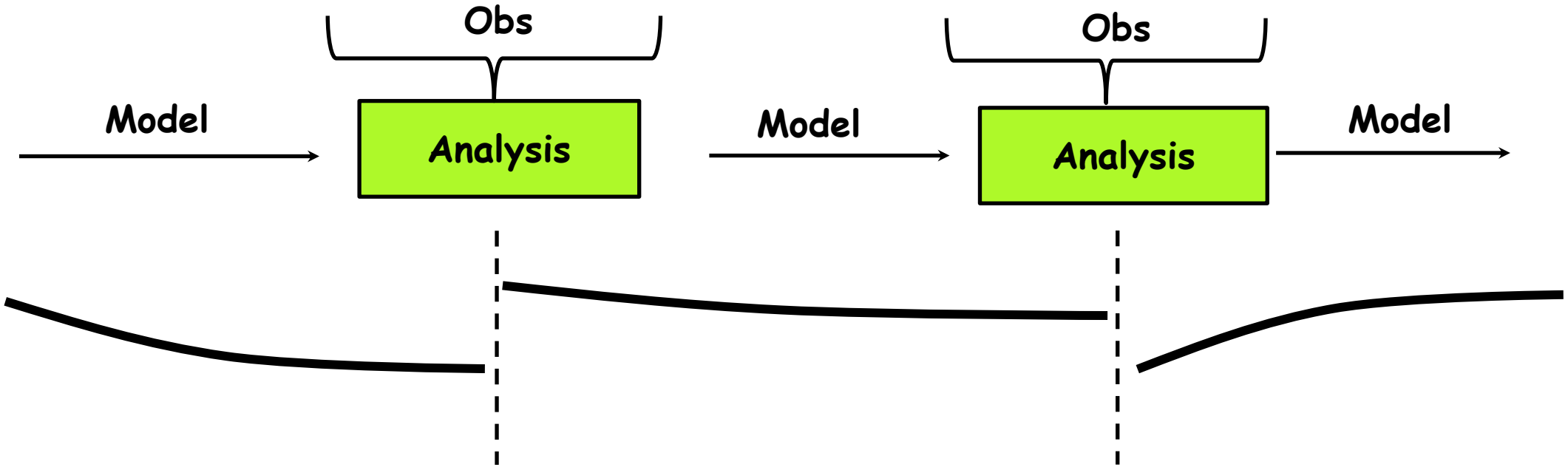
Intermittent

Observations processed in small batches

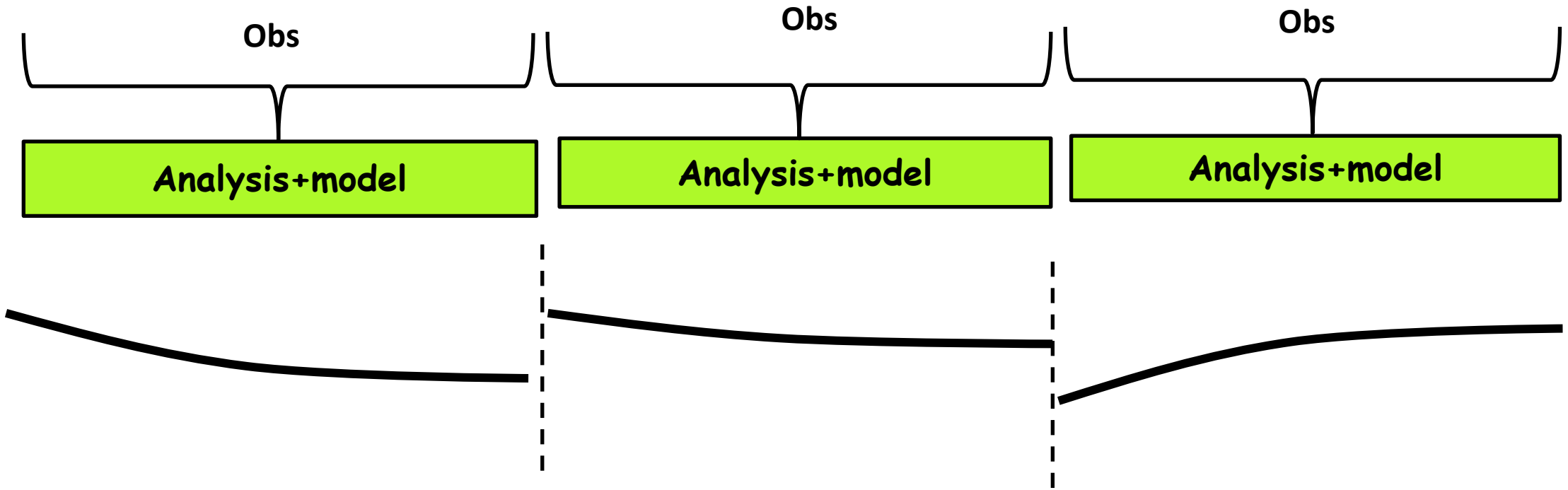
Continuous

Observation patches over long periods considered and the analysis correction is smoothed in time.

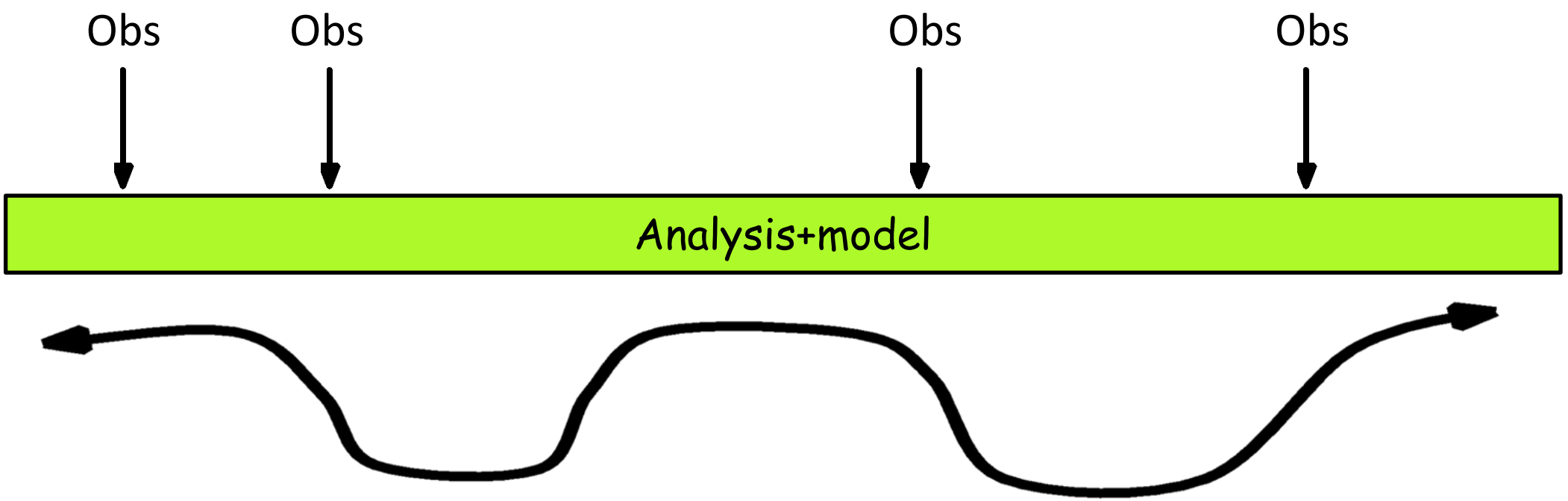
Sequential Intermittent Assimilation



Non-sequential Intermittent Assimilation

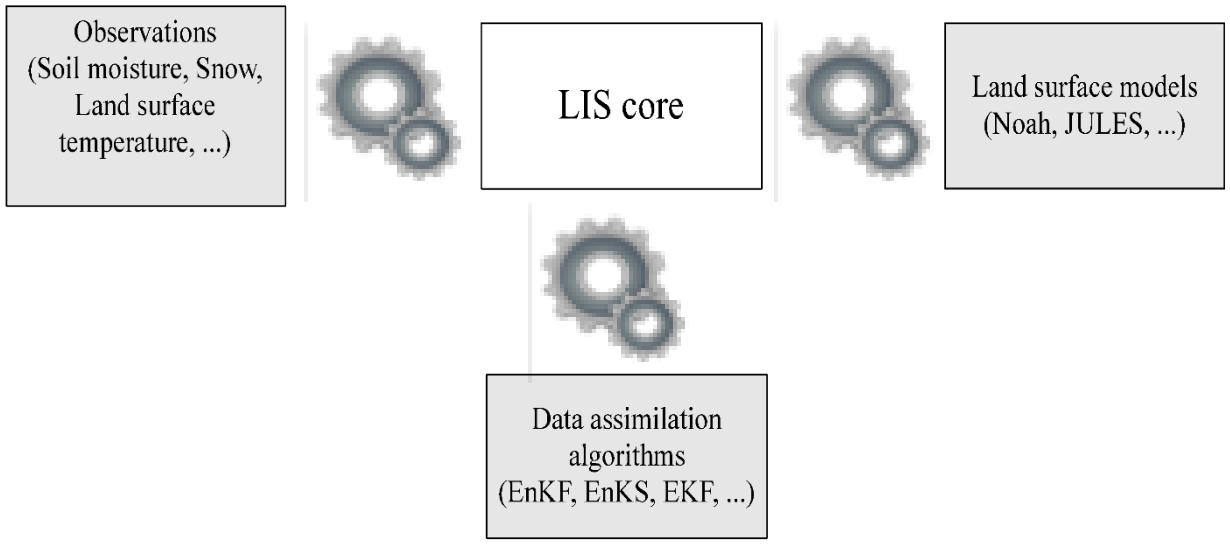


Non-sequential Continuous Assimilation



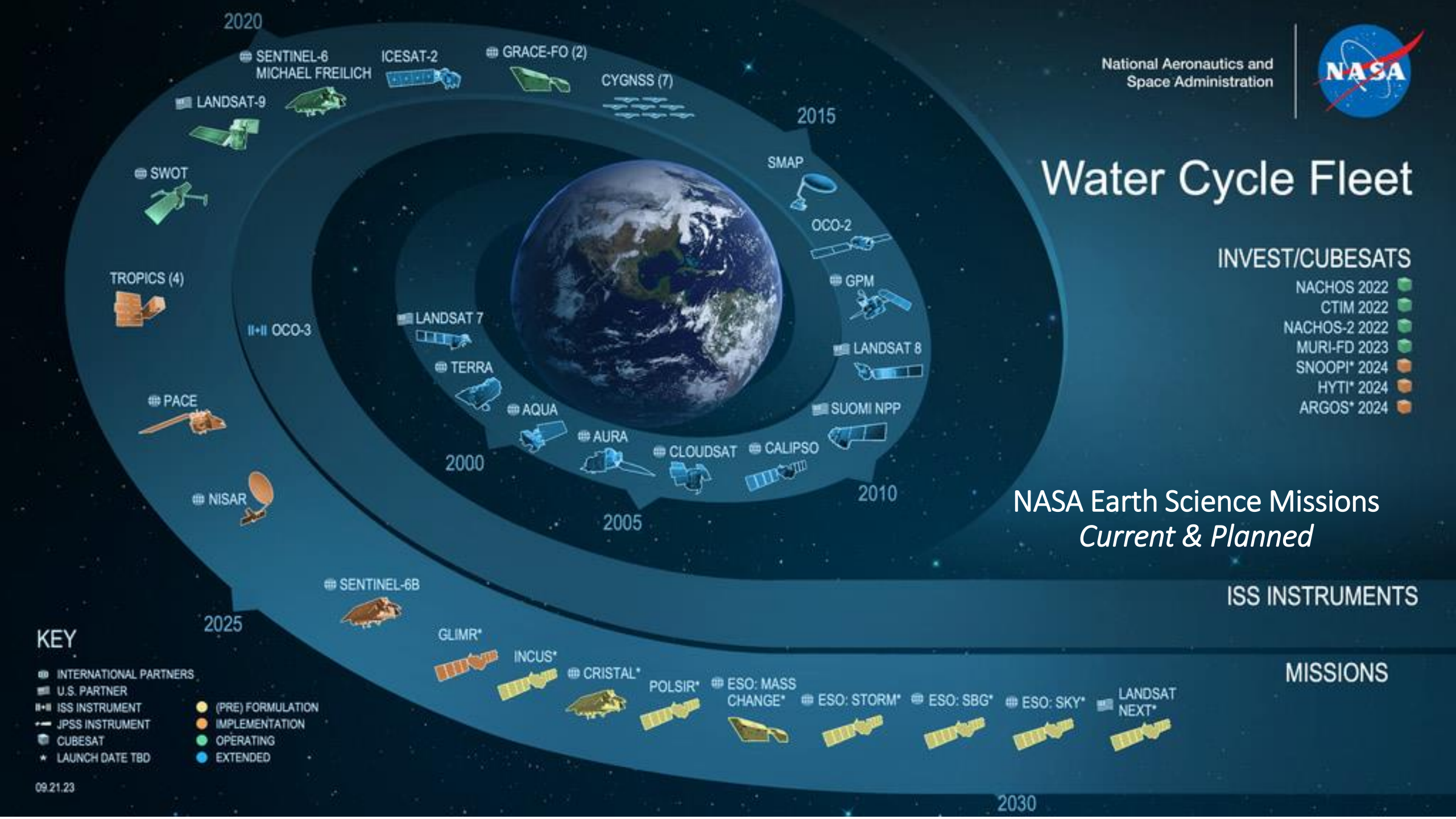
Data Assimilation subsystem in LIS

- Primarily used for state estimation - Corrects model states based on observations
- Advanced algorithms such as the Ensemble Kalman Filter (EnKF), Ensemble Kalman Smoother (EnKS) (originally developed based on the the NASA GMAO infrastructure)
- Supports the interoperable use of multiple land surface models, multiple algorithms and multiple observational data sources
- Support for concurrent data assimilation, forward models, radiance assimilation, observation operators employing advanced data fusion methods (deep learning)



Kumar, S.V., R.H. Reichle, C.D. Peters-Lidard, R.D. Koster, X. Zhan, W.T. Crow, J.B. Eylander, P.R. Houser (2008), A Land Surface Data Assimilation Framework using the Land Information System: Description and Applications, *Adv. Wat. Res.,* 31, 1419-1432.

Current/Future observations



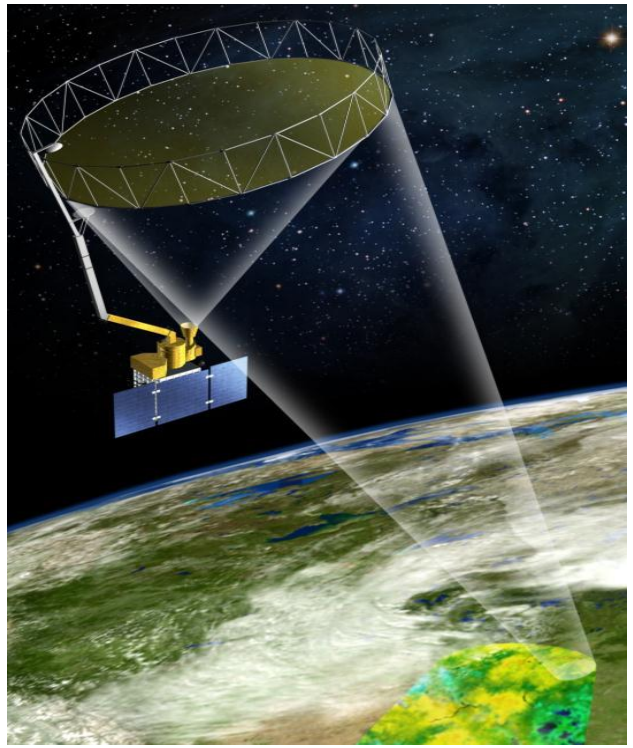
NASA Earth Science Missions
Current & Planned

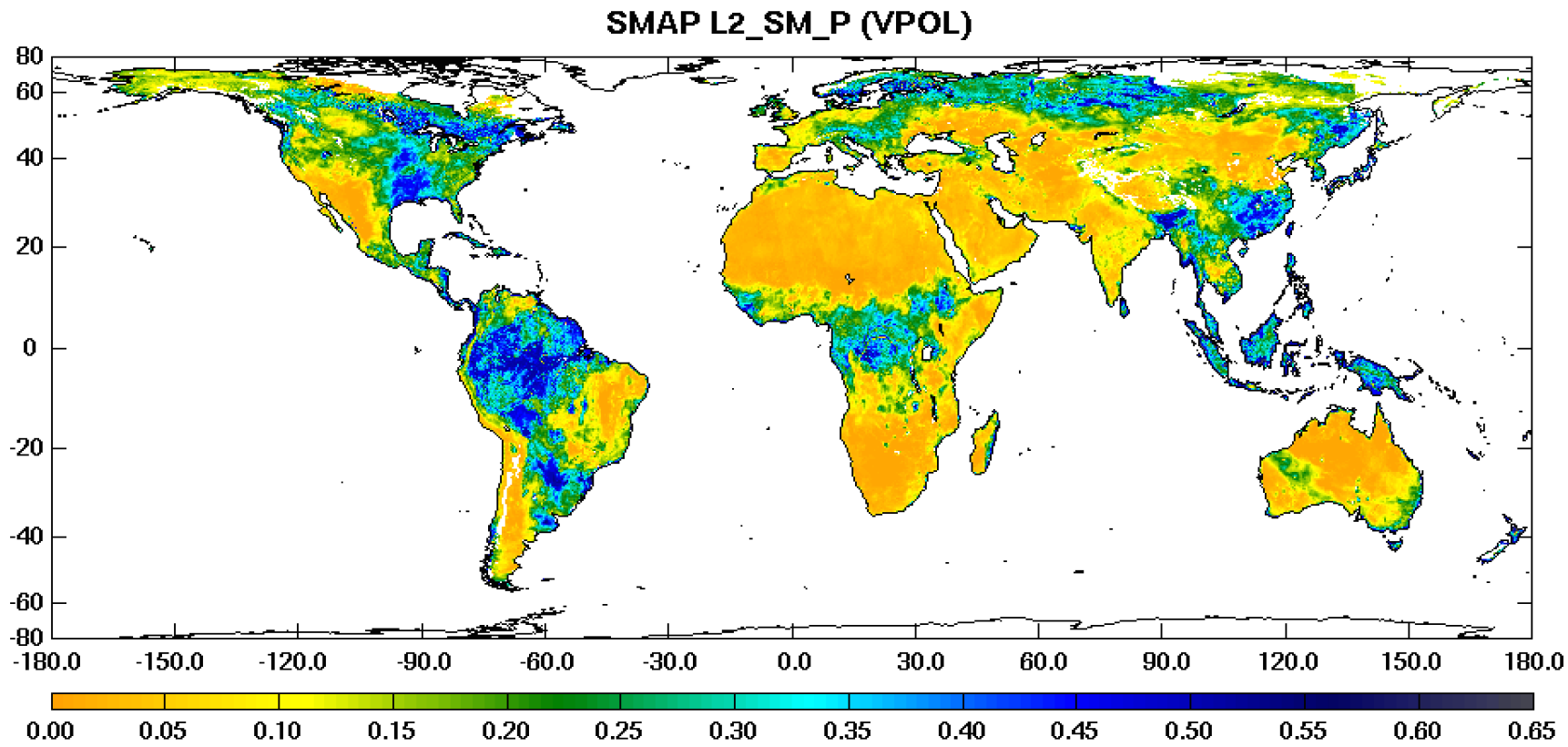
GPM and the IMERG Precipitation product



Precipitation from GPM core satellite, GPM constellation, and IMERG product (10km, 30 min)
Credit: GSFC Scientific Visualization Studio

Soil Moisture Active Passive (SMAP)





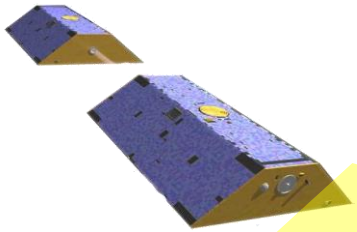
SMAP soil moisture (volumetric) based on radiometer data from May 25-27, 2015.
Credit: Peggy O'Neill/GSFC.

SMAP Facts

- **Resolution: 9km, daily**
- **Instruments: L-band Radiometer**
- **Launch: January 31, 2015**
- **Data: March 31, 2015-present**

SMAP Enhanced L3 Radiometer Global Daily 9 km EASE-Grid Soil Moisture, Version 1

GRACE and Trends in Terrestrial Water Storage

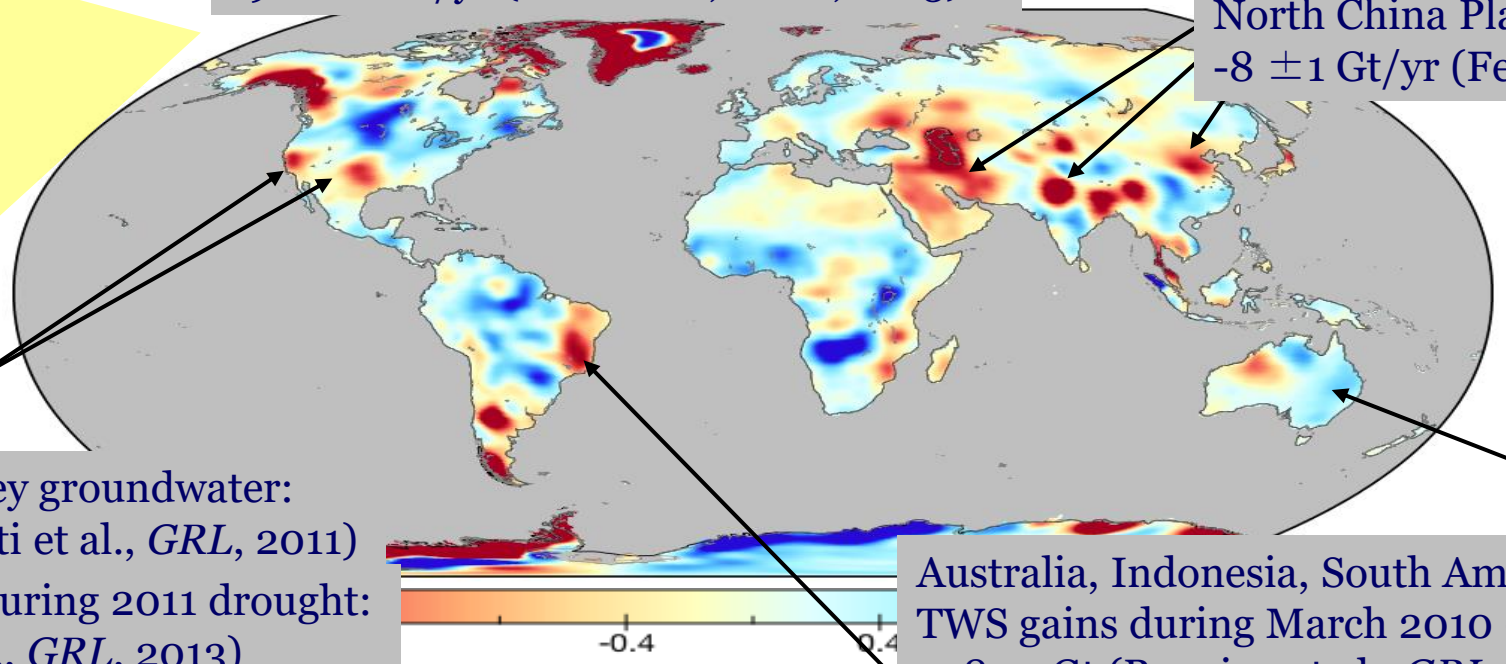


Iran TWS depletion:
-25 ±3 Gt/yr (Joodaki et al., *WRR*, 2014)

Groundwater depl

North-central Middle East groundwater:
-91 ±11 Gt/yr (Voss et al., *WRR*, 2013)

India groundwater depletion:
-54 ±9 Gt/yr (Tiwari et al., *GRL*, 2009)
North China Plain groundwater:
-8 ±1 Gt/yr (Feng et al., *WRR*, 2011)



Drought and groundwater depletion

California Central Valley groundwater:
-3 ±1 Gt/yr (Famiglietti et al., *GRL*, 2011)
Texas TWS depletion during 2011 drought:
-62 ±18 Gt (Long et al., *GRL*, 2013)

Progression from

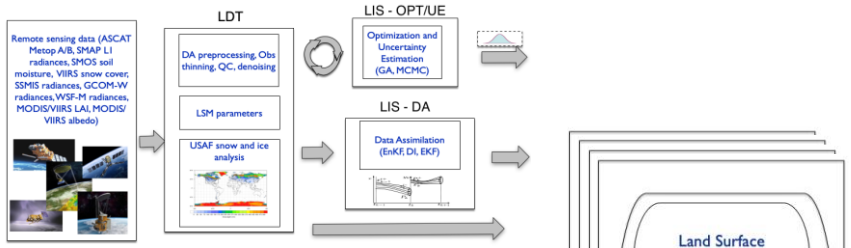
Australia, Indonesia, South America and Southeast Asia TWS gains during March 2010 –May 2011 La Nina:
-1,800 Gt (Boening et al., *GRL*, 2012)

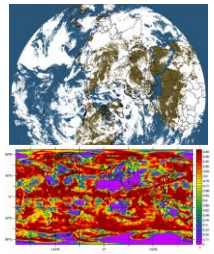
Eastern Brazil TWS losses during February 2012 – January 2015 drought:
-105 Gt/yr (Getirana, *J. Hydrometeor.*, 2013)

Drought

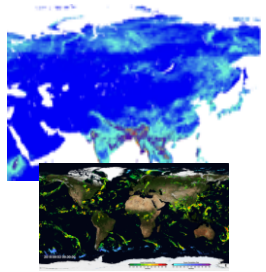
SWOT HyMAP DA framework

LIS Framework

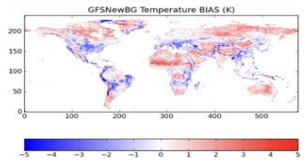




WWMCA COD to derive surface radiation analysis

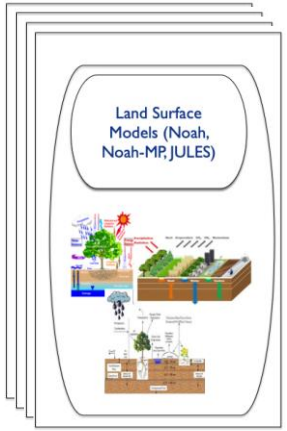


Surface obs, GEOPRECIP, SSM/I, CMORPH, IMERG,GALWEM merged to derive precipitation analysis using a bratseth scheme

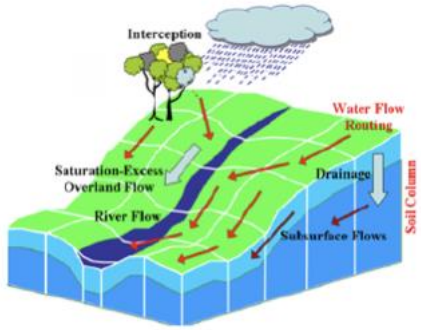


GALWEM/GFS first guess combined with obs of T, u, v, RH to derive shelter analysis using Bratseth

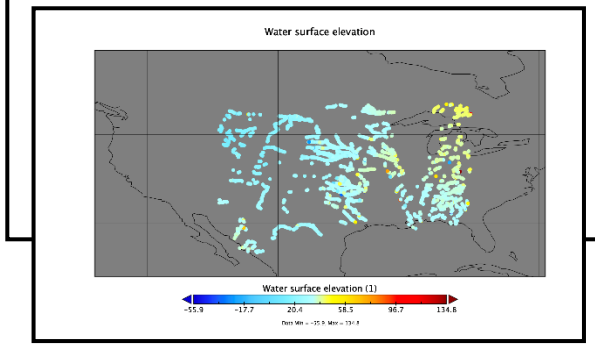
Meteorological analysis and inputs



HyMAP

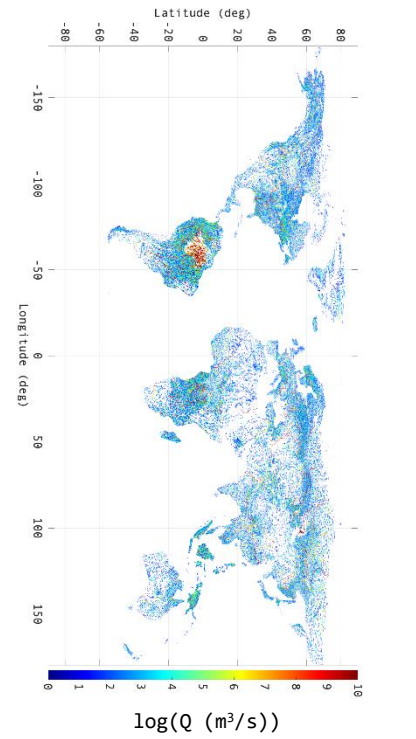


HyMAP-DA



Convert SWOT L2 reach shapefile to fit LIS GRID

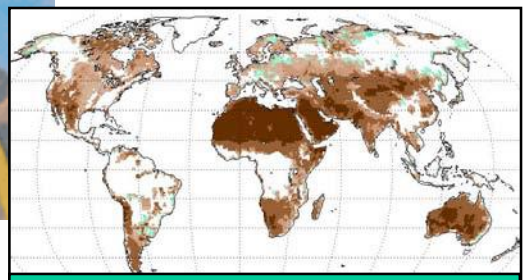
Streamflow



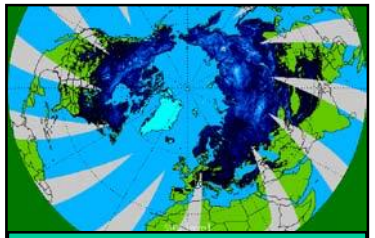
Current/Future Observations



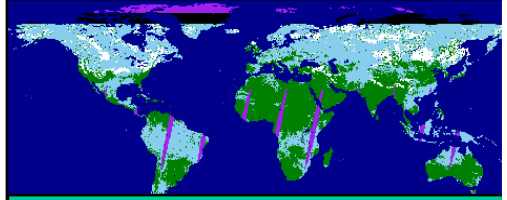
Vegetation/Carbon
(AVHRR, MODIS, LandSAT
DESDynI, ICESat-II, HyspIRI, LIST, ASCENDS)



Surface soil moisture
(SMMR, TRMM, AMSR-E, *SMOS, Aquarius, SMAP*)



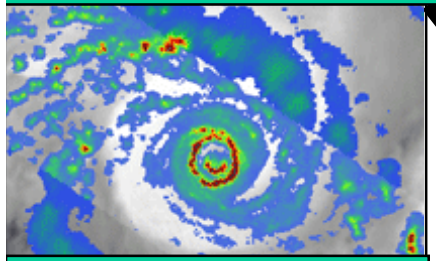
Snow water equivalent
(AMSR-E, SSM/I, *SCLP*)



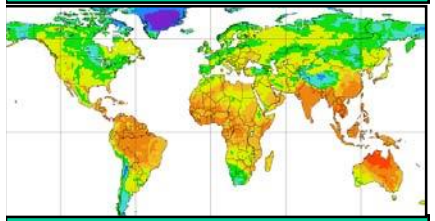
Snow cover fraction
(MODIS, *VIIRS, MIS*)



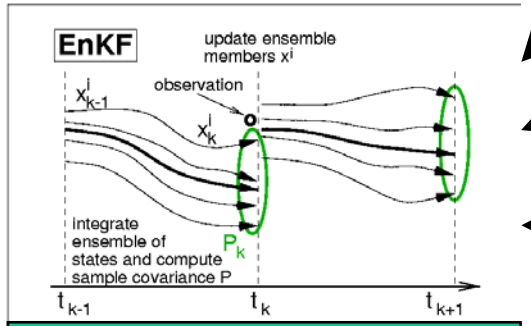
Water surface elevation
(*SWOT*)



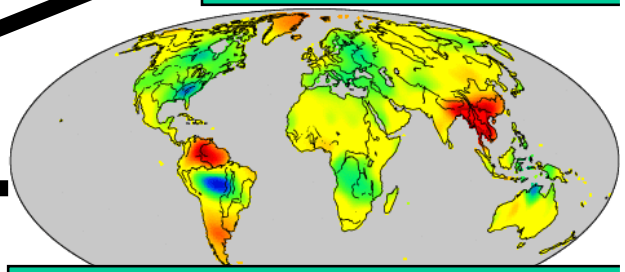
Precipitation
(TRMM, *GPM*)



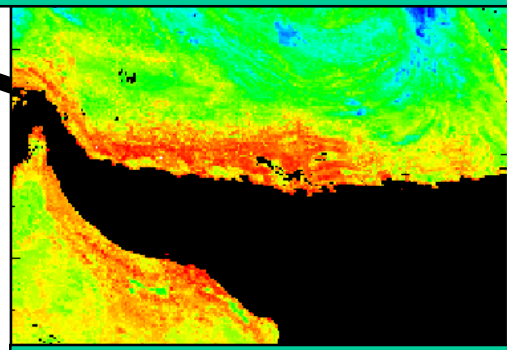
Radiation
(CERES, *CLARREO*)



Ensemble-based land data assimilation system



Terrestrial water storage (GRACE)



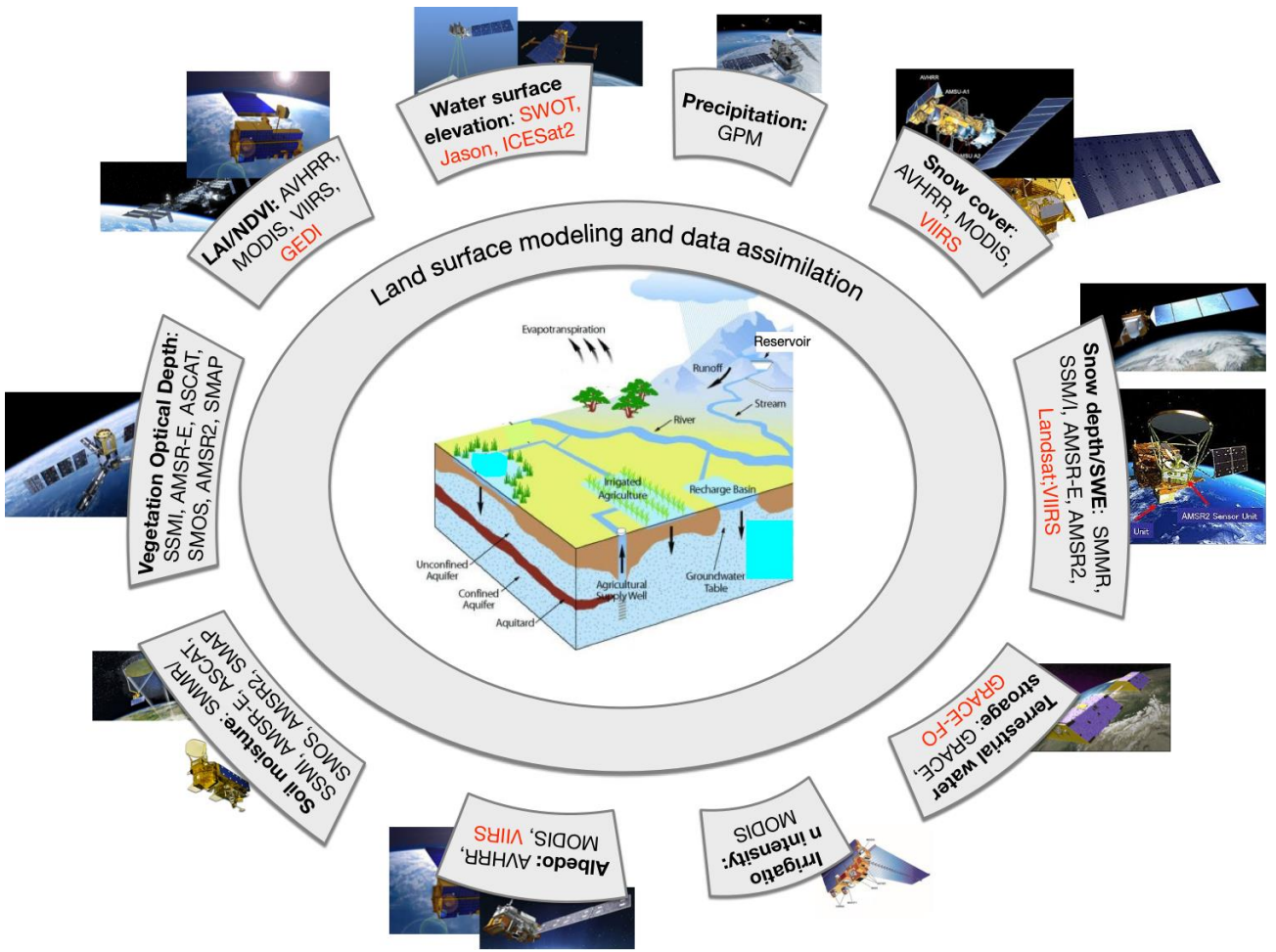
Land surface temperature
(MODIS, AVHRR,GOES,…)

Land surface data for research and applications:
Comprehensive view of land surface water/energy/carbon cycle.
Learn about processes, characterize errors, improve models.
Enhance weather and climate forecast skill.
Develop improved flood prediction and drought monitoring capability.
…

LIS-DA applications

LIS-DA applications

- Supports a large suite of variables and sensors.
- Assimilation is supported across multiple classes of models (LSM, routing, ..)
- Operational/Routine environments:
  - ☑ U.S. Air Force Weather (NWP, hydro, agricultural needs, …)
  - ☑ Famine early warning (FEWSNET)
  - ☑ GLDAS (drought monitoring)
  - ☑ NOAA NCEP (NWP, seasonal forecasting, drought monitoring)
  - ☑ NASA SPoRT (NWP, Flood forecasting)



Kumar, S.V., R.H. Reichle, C.D. Peters-Lidard, R.D. Koster, X. Zhan, W.T. Crow, J.B. Eylander, P.R. Houser (2008), A Land Surface Data Assimilation Framework using the Land Information System: Description and Applications, *Adv. Wat. Res.*, 31, 1419-1432.

Projects

Air Force / 557th Weather Wing

Find more information here regarding the 557 project (Restricted Access).

557th Weather Wing Website

NLDAS

NLDAS Website

GLDAS

GLDAS Website

NCA-LDAS

NCA-LDAS Website

NASA's Hydrological Forecast and Analysis System

NHyFAS - FEWSNET Hydrological Forecasts

USAID/FEWS NET

USAID/FEWS NET Website
FLDAS (FEWS NET Land Data Assimilation System) Website

SPoRT

NASA's Short-term Prediction Research and Transition (SPoRT) Center supports and maintains a real-time instance of LIS. Their operational version of LIS provides several applications and benefits to communities in the U.S., Africa, and other locations that contribute to enhancing assessments of drought and flood risk.
SPoRT Real-time Land Information System Blog Updates
SPoRT - Real-time 3km Land Information System (CONUS region)
Learn LIS! - Online Training Modules (provided by SPoRT)

NU-WRF

NASA-Unified Weather Research and Forecasting (NU-WRF)
NU-WRF Website

PMM

Precipitation Measurement Missions
PMM Website

US Air Force 557 Weather Wing

lis.gsfc.nasa.gov/projects/usaf/data

LIS Framework
Land Information System

About Docs Source Testcases Support Projects Pubs News

557WW Data

557 WW near real time runs with LIS-7.4

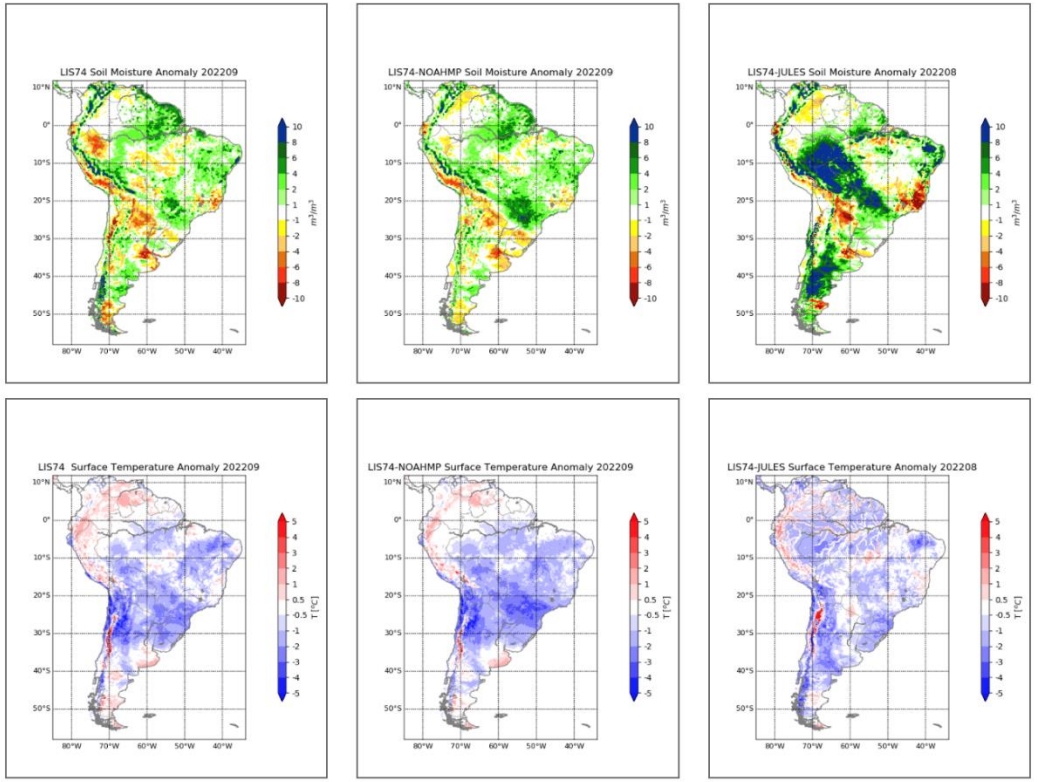
LIS Land surface model Noah-3.9 is run at 0.14-deg by 0.09-deg resolution on a global domain (latitude/longitude projection). Gridded snow depth analyses from SNODEP and soil moisture from SMAP and ASCAT are assimilated in the Noah-3.9 simulations. Gridded precipitation fields are generated using a Bratseth scheme (using NWP background field; CMORPH, GEOPRECIP, and SSMI/S satellite retrievals; and rain gauge reports). Gridded 2-m temperature, 2-m relative humidity, and 10-m wind speed fields are also generated by using a Bratseth scheme (using NWP background and surface weather reports). The Air Force GALWEM weather model is usually used for NWP background fields, with the NOAA GFS used as an emergency backup.

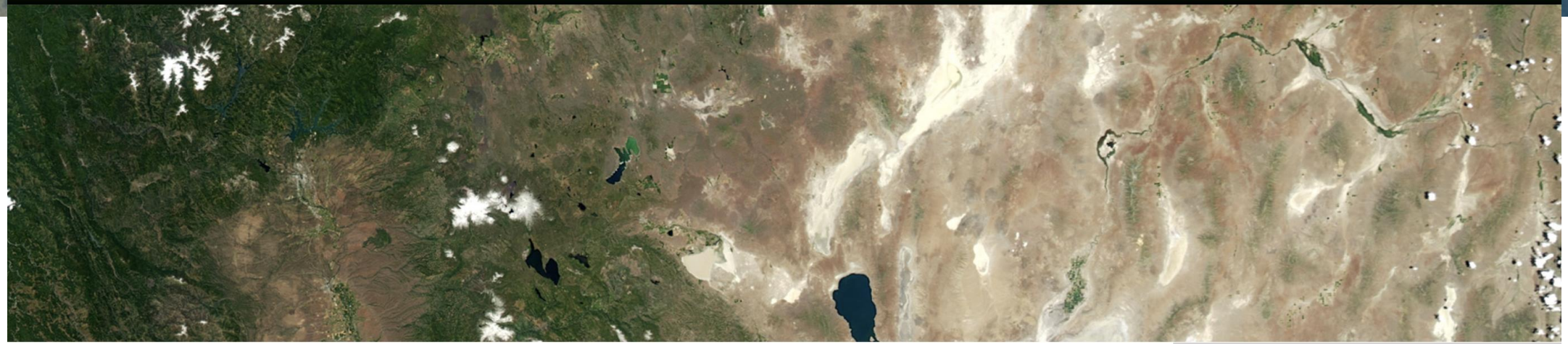
Kemp, E., J.W. Wegiel, S.V. Kumar, J. V. Geiger, D.M. Mocko, J. Jacob, C.D. Peters-Lidard, 2022: A NASA-Air Force Precipitation Analysis for Near-Real-Time Operations, *Journal of Hydrometeorology,* 23(6), 965-989, doi:10.1175/JHM-D-21-0228.1

Displaying Items 1 - 8 of 8

Product Type: - Any - Region: South America Variable/Metric: - Any - Apply

LIS74 Anomalies: South America





Land Data Assimilation System

Land Data Assimilation Systems (LDAS) aim to produce high quality fields of land surface states (e.g., soil moisture, temperature) and fluxes (e.g., evapotranspiration, runoff) by integrating satellite- and ground-based observational data products, using advanced land surface modeling and data assimilation techniques.

The Hydrological Sciences Laboratory at NASA's Goddard Space Flight Center has been developing multiple LDAS and distributing LDAS output for more than two decades.

Motivation

- The land-surface component of the water cycle affects atmospheric and climatic processes while also controlling the distribution of freshwater, which is vital to life on land.
- Proper characterization of spatial and temporal variations in water and energy states (e.g., soil moisture and temperature) and fluxes (e.g., evaporation and runoff) and is critical to many scientific and practical applications, including weather prediction, agricultural forecasting, drought and flood risk assessments, and improving understanding of land-atmosphere interactions and climate change impacts.
- Numerous ground and space based observations are relevant to the water and energy cycles, but often they include errors or spatial

ntain exactly the information we need (for example, snow cover observations do not tell us

Get the Data

LDAS datasets are available from the NASA GES DISC

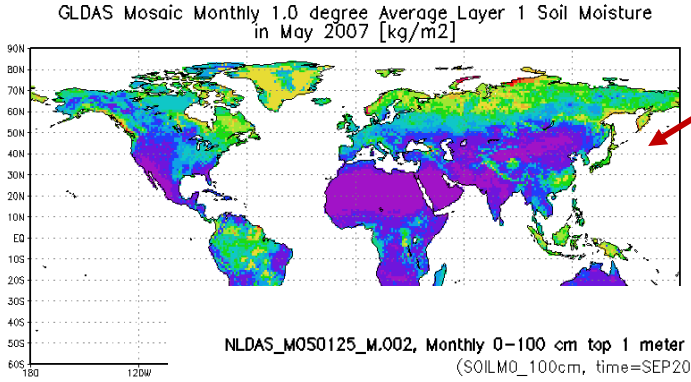


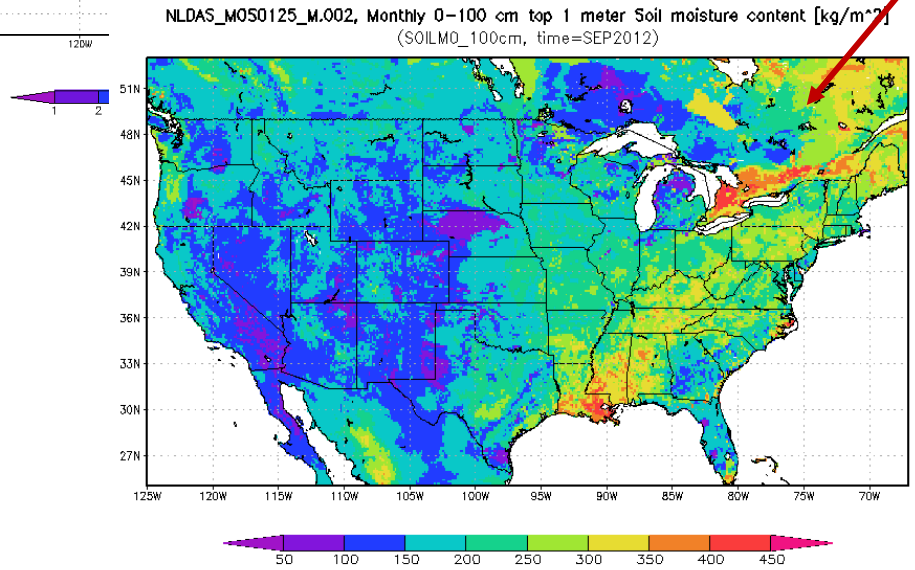
The NASA Goddard Earth Sciences Data and Information Services Center (GES DISC) provides access to LDAS datasets using multiple methods, including HTTPS, GDS, and the ability to subset spatially, temporally, and/or by variable:

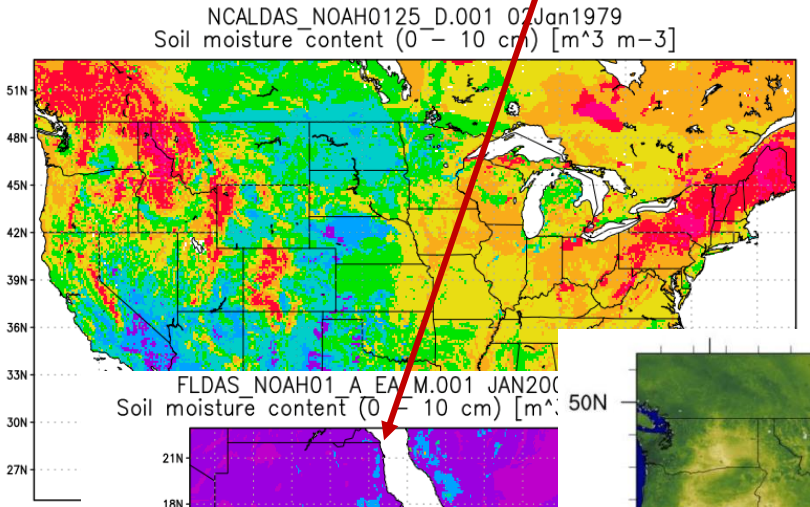
https://ldas.gsfc.nasa.gov/

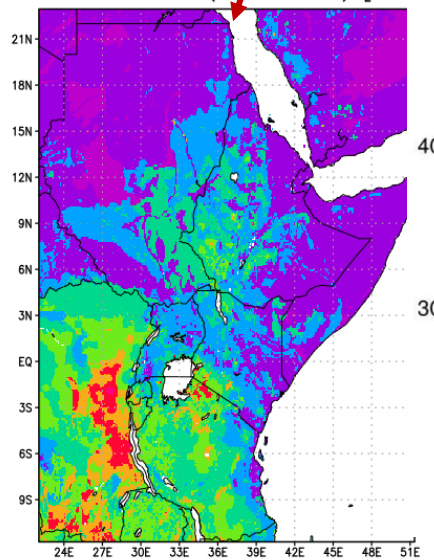
LDAS
Land Data Assimilation System

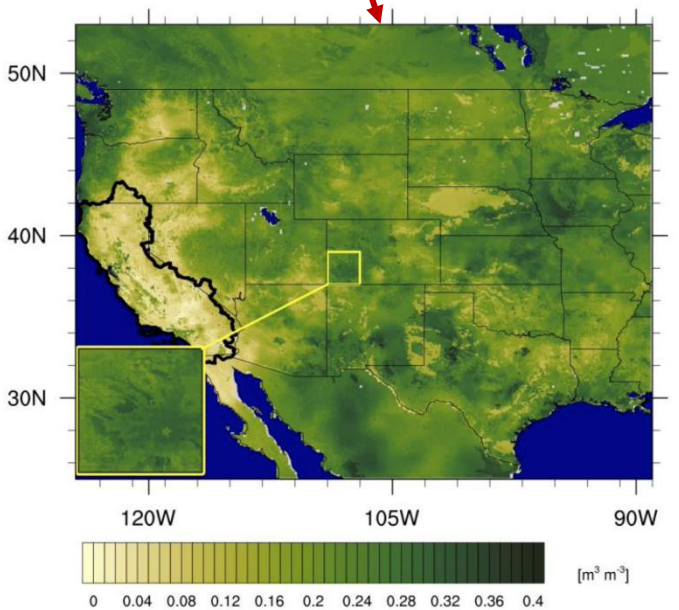
GLDAS ▾ NLDAS ▾ NCA-LDAS ▾ FLDAS ▾ WLDAS ▾ FAQ ▾ Get Data









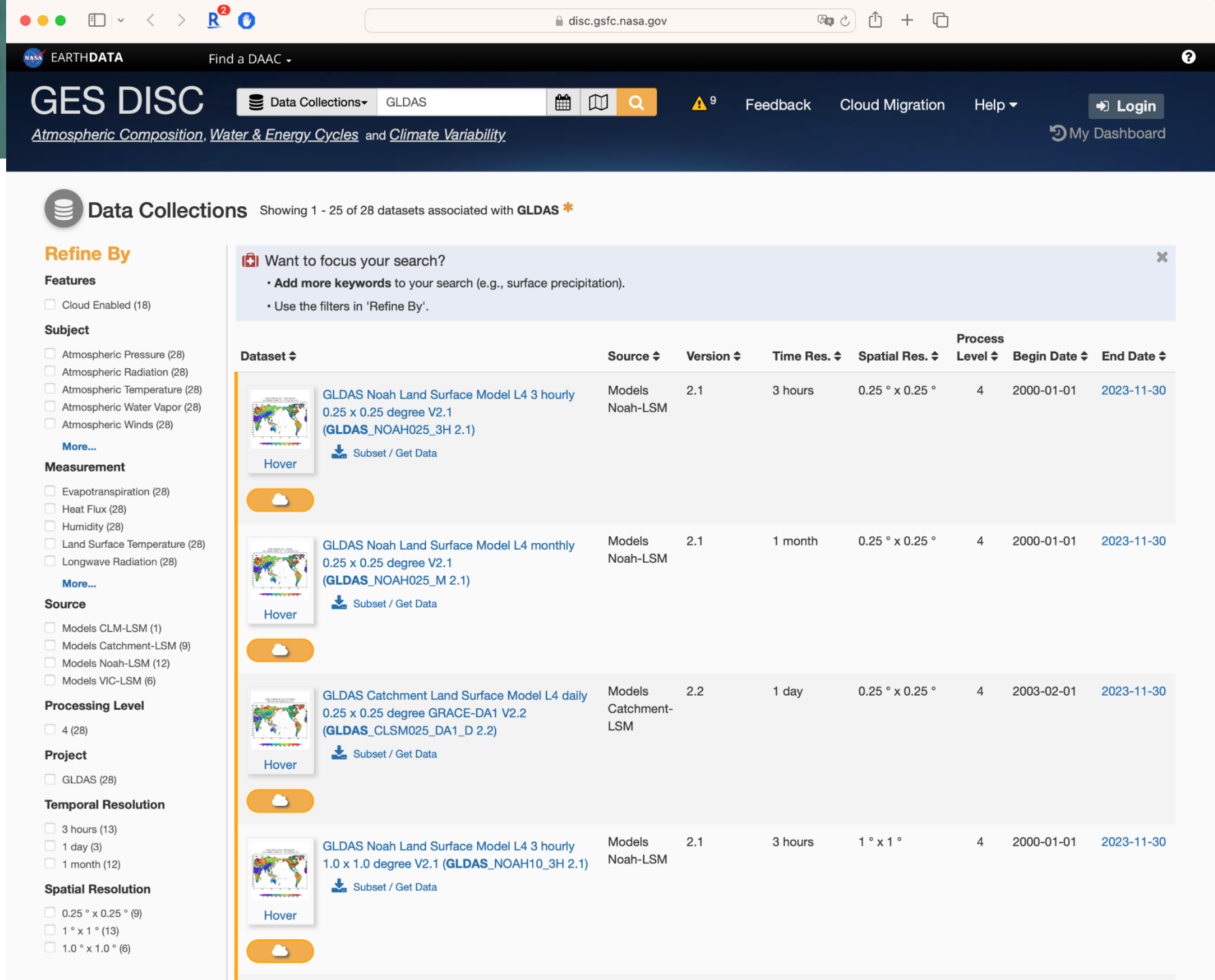


GODDARD EARTH SCIENCES

GODDARD EARTH SCIENCES

LDAS datasets are available from the **NASA GES DISC**

https://disc.gsfc.nasa.gov



GODDARD EARTH SCIENCES

LIS-DA configuration

Create the DA obs domain using LDT

Use LDT in the ‘DA preprocessing’ mode to generate the obs domain and scaling parameters

LDT run domain should reflect the intended observation grid (projection and resolution) – Note that this can be different from the model resolution and projection

A successful completion will generate a ‘domain’ file: lsm_cdf.nc

This will be used as input for the LIS run

```
LDT running mode:                "DA preprocessing"
```

```
Map projection of the LIS domain:   "ease V2"
Run domain lower left lat:           32.0
Run domain lower left lon:          -104.0
Run domain upper right lat:          43.0
Run domain upper right lon:         -93.0
Run domain grid designation:         "M09"
```

```
DA observation source:               "LIS LSM soil moisture"
DA preprocessing method:             "CDF generation"
Name of the preprocessed DA file:    'lsm_cdf'
Temporal resolution of CDFs:         "yearly"
Number of bins to use in the CDF:    100
Temporal averaging interval:         "1da"
Apply external mask:                 0
External mask directory:             none
Observation count threshold:         50
```

Creating the CDF files for a priori bias correction

Use LDT in the 'DA preprocessing' mode to generate the obs domain and scaling parameters

Within the DA preprocessing mode, three options are supported:

"Obs grid generation"

"CDF generation"

"Anomaly correction"

Temporal resolution (yearly/monthly)

To increase sampling density:

'Enable spatial sampling for CDF calculations:'

'Sampling window radius for CDF calculations:'

```
LDT running mode:                "DA preprocessing"
```

```
DA observation source:              "NASA SMAP soil moisture"
DA preprocessing method:            "CDF generation"
Name of the preprocessed DA file:   'smap_cdf'
Temporal resolution of CDFs:        "yearly"
Number of bins to use in the CDF:   100
Temporal averaging interval:        "1da"
Apply external mask:                0
External mask directory:            none
Observation count threshold:        50

NASA SMAP soil moisture observation directory: ../SMAP/SPL3SMP_E/
NASA SMAP soil moisture data designation:      SPL3SMP_E
```

Creating the CDF files for a priori bias correction

Two separate LDT runs should be conducted to generate the **model CDF** and the **observation CDF**

These files will be used as inputs to the LIS run

```
LDT running mode:                "DA preprocessing"
```

```
DA observation source:              "NASA SMAP soil moisture"
DA preprocessing method:            "CDF generation"
Name of the preprocessed DA file:   'smap_cdf'
Temporal resolution of CDFs:        "yearly"
Number of bins to use in the CDF:   100
Temporal averaging interval:        "1da"
Apply external mask:                0
External mask directory:            none
Observation count threshold:        50

NASA SMAP soil moisture observation directory: ../SMAP/SPL3SMP_E/
NASA SMAP soil moisture data designation:      SPL3SMP_E
```

```
DA observation source:              "LIS LSM soil moisture"
DA preprocessing method:            "CDF generation"
Name of the preprocessed DA file:   'lsm_cdf'
Temporal resolution of CDFs:        "yearly"
Number of bins to use in the CDF:   100
Temporal averaging interval:        "1da"
Apply external mask:                0
External mask directory:            none
Observation count threshold:        50

LIS soil moisture output model name:        "Noah.3.3"
LIS soil moisture output format:            "netcdf"
LIS soil moisture output methodology:       "2d gridspace"
LIS soil moisture output naming style:      "3 level hierarchy"
LIS soil moisture output map projection:    "latlon"
LIS soil moisture output nest index:        1
LIS soil moisture output directory:         ../DEF/OUTPUT
LIS soil moisture domain lower left lat:    33.0625
LIS soil moisture domain lower left lon:    -97.9375
LIS soil moisture domain upper right lat:   33.1875
LIS soil moisture domain upper right lon:   -97.8125
LIS soil moisture domain resolution (dx):   0.125
LIS soil moisture domain resolution (dy):   0.125
```

DA specifications in LIS

```
Number of data assimilation instances:            1
Data assimilation algorithm:                      EnKF
Data assimilation set:                            "SMAP(NASA) soil moisture"
Data assimilation exclude analysis increments:    0
Data assimilation number of observation types:    1

Data assimilation output interval for diagnostics:  "1da"
Data assimilation output processed observations:    1
Data assimilation output ensemble spread:           0
Data assimilation scaling strategy:                 "CDF matching"
Data assimilation output innovations:               0

Data assimilation use a trained forward model: 0
Data assimilation trained forward model output file: none

Bias estimation algorithm:                        "none"
```

Assimilation algorithm: EnKF

Assimilation Set: Observation to be assimilated

Options to output processed obs, ensemble spread, innovations

```
Number of data assimilation instances:            3
Data assimilation algorithm:                      "EnKS" "EnKF" "EnKF"
Data assimilation set:                            "GRACE TWS" "AMSR-E(LPRM) soil moisture" "ANSA snow depth"
Data assimilation exclude analysis increments:    0 0 0
Data assimilation number of observation types:    1 1 1

Data assimilation output interval for diagnostics:  "1da" "1da" "1da"
Data assimilation output ensemble members:          0 0 0
Data assimilation output processed observations:    0 1 1
Data assimilation output innovations:               0 0 0
```

Assimilation algorithm: EnKS, EnKF, EnKF

If multiple data assimilation instances are chosen, then they are specified as additional columns.

Specifying perturbations

```
Apply perturbation bias correction:    1
Forcing perturbation algorithm:          "GMAO scheme"
Forcing perturbation frequency:          "1hr"
Forcing attributes file:                 ./forcing_attribs.txt
Forcing perturbation attributes file:    ./forcing_pert_attribs.txt
```

forcing_attribs.txt
specifies the variables and their ranges

```
#varmin  varmax
Incident Shortwave Radiation Level 001
0.      1300.
Incident Longwave Radiation Level 001
-50.     800.
Rainfall Rate Level 001
0.0      0.001
```

forcing_pert_attribs.txt
specifies the perturbation settings

```
#ptype   std    std_max   zeromean  tcorr  xcorr ycorr ccorr
Incident Shortwave Radiation Level 001
1  0.30     2.5    1        86400  0     0     1.0  -0.5  -0.8
Incident Longwave Radiation Level 001
0  50.0     2.5    1        86400  0     0    -0.5   1.0  0.5
Rainfall Rate Level 001
1  0.50     2.5    1        86400  0     0    -0.8   0.5  1.0
```

Perturbation type: additive (0) or multiplicative (1)

Std: standard deviation of perturbations

Zero mean: enforce zero mean across the ensemble

Std_max: maximum allowed normalized perturbation (relative to $N(0,1)$)

Tcorr: temporal correlation scale (in seconds) used in the AR(1) model

Xcorr, Y-corr: Spatial correlation scale (deg)

Ccorr: cross correlations between variables

State perturbations

State perturbation algorithm: "GMAO scheme"
State perturbation frequency: "3hr"
State attributes file: ./noah_sm_attribs.txt
State perturbation attributes file: ./noah_sm_pertattribs.txt

noah_sm_attribs.txt

specifies the variables and their ranges

```
#nfields
4
#name  varmin  varmax
Soil Moisture Layer 1
  0.01       0.55
Soil Moisture Layer 2
  0.01       0.55
Soil Moisture Layer 3
  0.01       0.55
Soil Moisture Layer 4
  0.01       0.55
```

noah_sm_pertattribs.txt

specifies the perturbation settings

```
#perttype  std    std_max   zeromean  tcorr  xcorr ycorr ccorr
Soil Moisture Layer 1
  0  0.001     2.5          1       43200   0     0    1.0 0.6 0.4 0.2
Soil Moisture Layer 2
  0  0.00      2.5          1       43200   0     0    0.6 1.0 0.6 0.4
Soil Moisture Layer 3
  0  0.00      2.5          1       43200   0     0    0.4 0.6 1.0 0.6
Soil Moisture Layer 4
  0  0.00      2.5          1       43200   0     0    0.2 0.4 0.6 1.0
```

Global Hydro Intelligence environment

Global Hydro Intelligence (GHI)

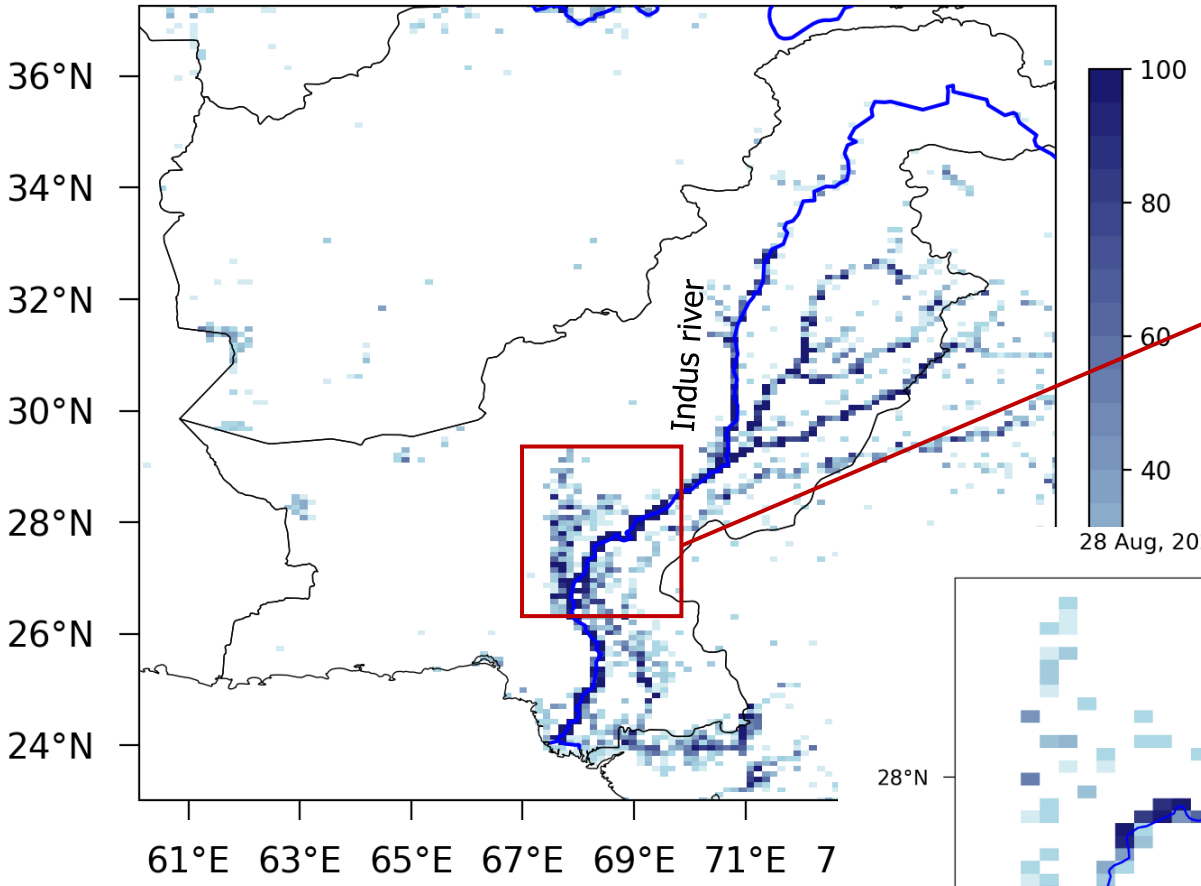
A Comprehensive Framework for hydrologic analysis, forecasting, and projections across time scales



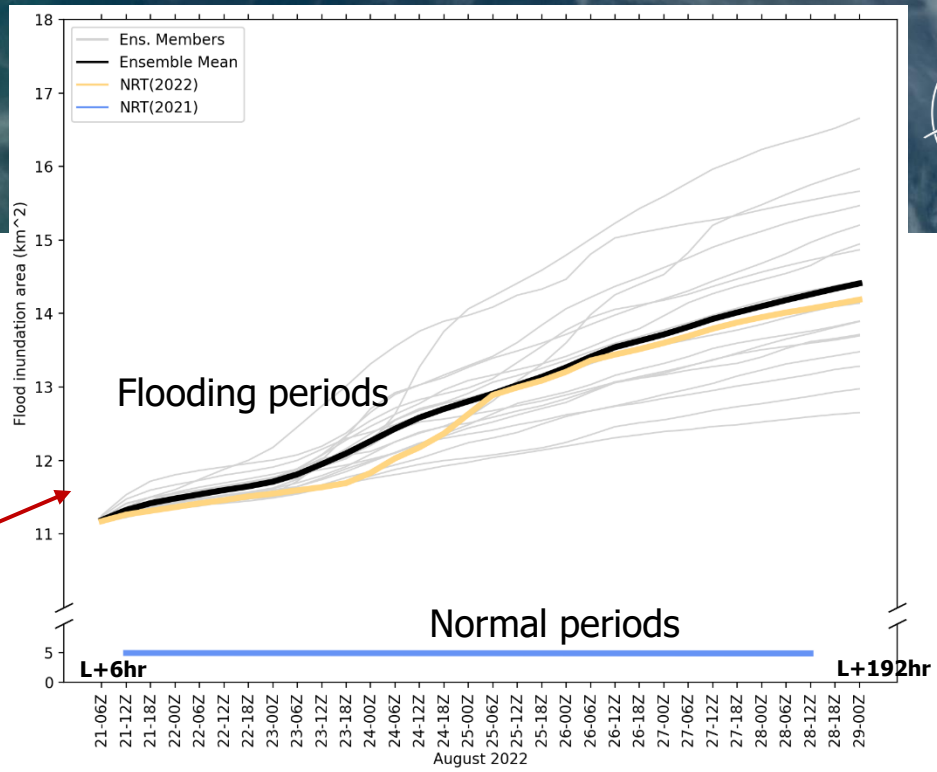
NRT | Medium-range (0-16 days) | S2S (0-9 months) | Interannual (up to 100 years)

GHI-Medium Range (MR) forecast : Pakistan Flood of August 2022

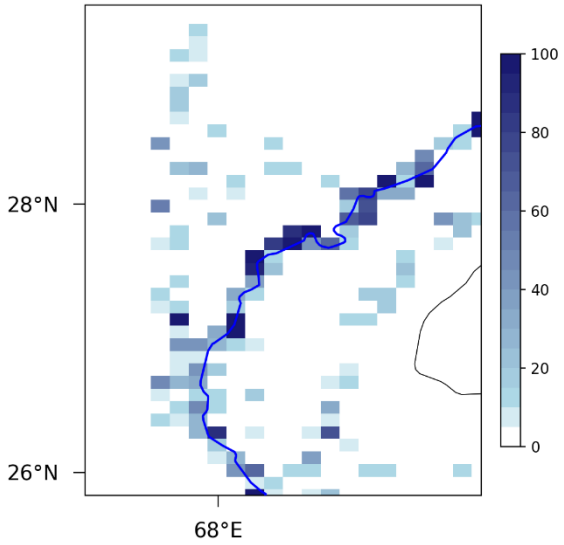
Flood inundation area (km^2): 28 Aug, 2022 (Lead+192hr)

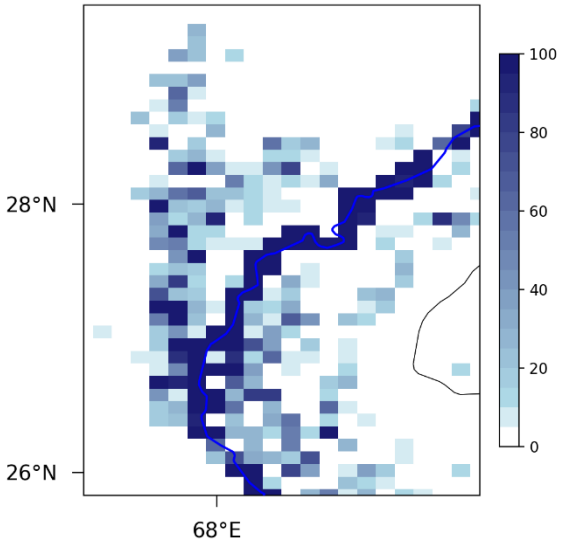


28 Aug, 2021



28 Aug, 2022 (Lead+192hr)



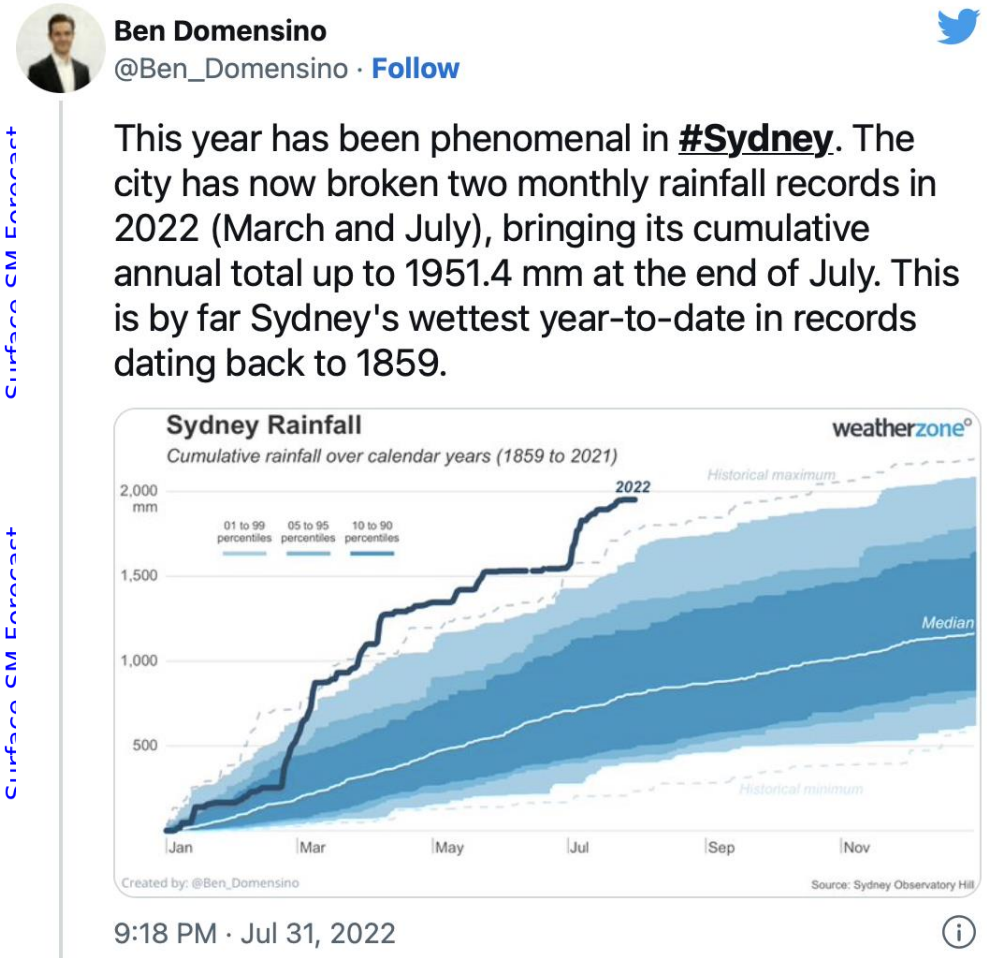


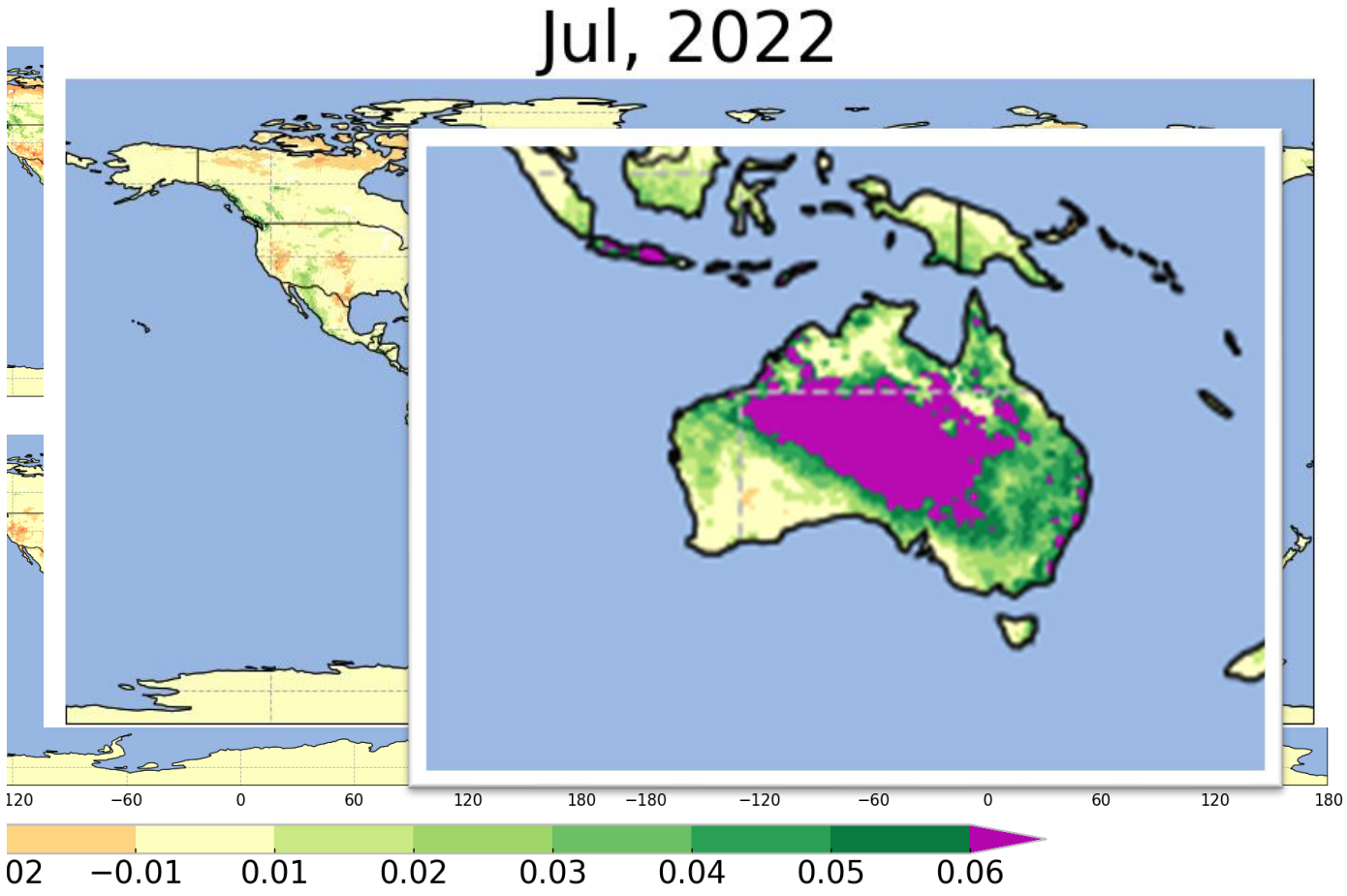


MODIS (August 28,2022)

GHI Subseasonal-to-Seasonal (S2S) Forecast : Extreme Wet Forecasts for Australia

May-2022 forecast, lead = 2, July





https://www.theguardian.com/australia-news/2022/aug/17/eastern-australia-faces-wet-weather-and-flooding-with-70-chance-of-third-consecutive-la-nina

[1] Contains modified Copernicus Sentinel data, 2023, from the C3S Climate Bulletin, 9 Aug, 2023, https://en.wikipedia.org/wiki/2023_heat_waves#/media/File:July_2023_was_the_earmest_globally.png

Top Ref.: Tunisia's National Institute of Meteorology's May- and June-2023 reports: https://www.meteo.tn/en/actualites/climatological-report-month-may-2023-tunisia

GHI Inter-annual (IA) : Future Climate assessments

Significant increases in extreme hot days are projected at Al Udeid AB, Qatar with more than three months in a year experiencing extreme hot days.

This, in combination with other health and environmental factors infers significant investments will be required to sustain operations at this location.

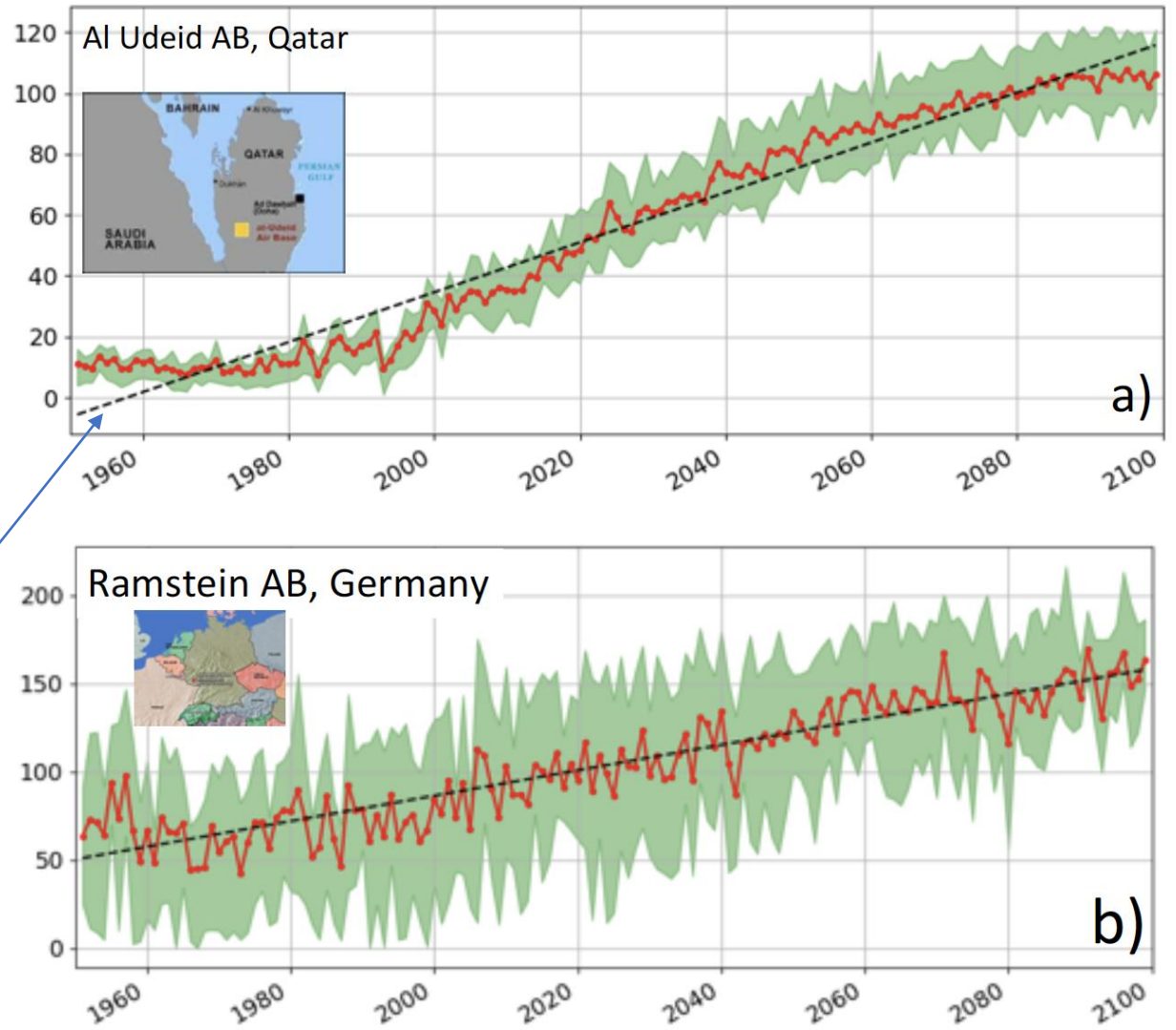


Figure 2. Panel a) Number of days per year when land surface temperatures are above 90th percentile (analog of extreme heat days), based on SSP2-4.5. Panel b) Number of days per year when soil moisture is below 10th percentile (increases in dry extremes), also based on SSP2-4.5. Green shadings show the spread across 25 CMIP6 ensembles (25-75 percentiles); red lines are the ensemble averages; dashed black lines indicate the linear trends that are statistically significant.